

Factoring out the greatest common factor (GCF)

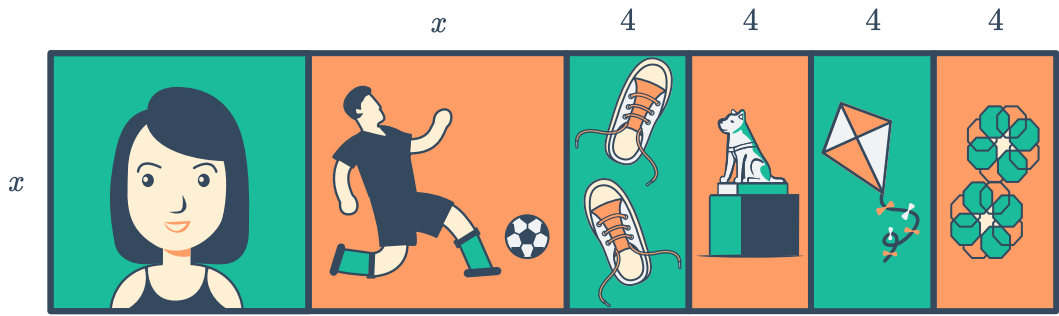


- 1
- a 3 b a c $3x$ d $3b$
- 2
- a $4r + 12 = 4(r + 3)$ b $5v - 30 = 5(v - 6)$
c $18c - 24 = 6(3c - 4)$ d $2t^2 + 2t = 2t(t + 1)$
- 3
- a $6(v + 5)$ b $5(9t - 8)$ c $-2(s + 5)$ d $-2(6s - 5)$
e $y(y + 4)$ f $k(k - 5)$ g $2u(u - 4)$ h $2t(2 + t)$
i $x(42 - x)$ j $7(x + 5)$ k $6(6 - x)$ l $-2(x - 5)$
m $-10(10 + x)$ n $-6(3y^2 - 1)$ o $13(3x + 4)$ p $(t - 3)(8t + 9)$

Additional questions

- 4 When factoring $5xy$ out of the term $-5xy$ you get $\frac{-5xy}{5xy} = -1$ but Xander eliminated the term instead.
- 5
- a Let the width of the smallest field be x so that its area is x^2 . The largest field has an area of $9x^2$ and the middle field has an area of $x(x + 5)$.
- b The area of the entire field is the sum of each smaller field: $x^2 + 9x^2 + x(x + 5)$. The factored form of this expression is $x(11x + 5)$ so the width of the field is x and the length is $11x + 5$
- 6 When the greatest common factor is a monomial, we can use monomial division to determine what expression is left after the factoring is complete.
- 7 The greatest common factor represents the largest value that can be divided evenly from each term. If the greatest common factor of a variable had an exponent larger than any of the original terms, then dividing it out would leave a negative exponent. For example, consider the expression $x^2 + x^3$. If we divide both terms by x^2 (the smaller exponent) then we are left with $1 + x$. If instead we try to divide both terms by x^3 (the larger exponent) then we would be left with $x^{-1} + 1$.

a $x(2x + 16)$



b $2x(x + 8)$

